Contribution Journal

ITAI 1371 Machine Learning

Midterm Project: Stroke Risk Prediction

Dataset	healthcare_disease_prediction_2000.csv (2,000 rows); target: Stroke
GitHub repository	ITAI-1371-ML / Midterm Project
Submission date	10/28/2026
Team	Camila, Danh, Great, Karima, Quan

Each member completes their own section below, in their own words. Describe what you personally did, which cells, files or documents you produced, and what you learned. This journal is used to assign each member's share of the grade, so be specific.

1. Team Summary

Fill in after all members have completed their sections. Contribution % should total 100.

Member	Role	Main deliverables	Contribution
Karima	Data & Split Lead	Original dataset link/source document; data loading and 70/30 split script; missing-values inspection report; saved train and test CSV splits	18%
Danh	EDA Lead (training set only)	Comprehensive EDA notebook section with distribution plots; key medical risk insights passed to Members 3 and 4	18%
Great	Encoding, Scaling & Normalization Lead	Encoding and scaling code section; before/after feature scaling comparison plots; sample transformed rows output	23%
Quan	Feature Engineering & Balancing Lead	Feature engineering code and rationale; before/after class balance plot; final cleaned and balanced training dataset export	23%
Camila	Documentation & GitHub Lead	Complete GitHub repository; 1-page proposal PDF and reflection journal; member contribution log; final clean dataset (.csv) and Jupyter notebook (.ipynb)	18%

2. Karima — Data & Split Lead

Assigned responsibilities

- Document the original data source URL and load healthcare_disease_prediction_2000.csv in Jupyter
- Print .head(), .shape and .info() to verify 2,000 rows and check data types
- Perform the 70% training / 30% testing split in Python (train_test_split)
- Handle initial missing-value imputation (fitted on the training set only)

Assigned deliverables

Original dataset link/source document; data loading and 70/30 split script; missing-values inspection report; saved train and test CSV splits

What I actually did

For our midterm project, I contributed to preparing the healthcare disease prediction dataset for machine learning analysis. We worked with a dataset containing 2,000 records and divided it into 70% training data and 30% testing data using Python's train_test_split.

One part I found difficult was understanding how to split the dataset correctly and how the training and testing sets are used in machine learning. I also found it challenging to understand the Python code and how each step connects to the next. Checking missing values and understanding the missing-value report took extra effort because I was still learning how to interpret the results.

Another difficult part was understanding why exploratory data analysis (EDA) should be performed only on the training set. Learning about data leakage was also challenging for me, especially understanding why an imputer must be fitted on the training data only and then used to transform the testing data. Even though some steps were difficult, I tried to understand their purpose and how they contribute to preparing reliable data for a machine learning model.

3. Danh — EDA Lead (training set only)

Assigned responsibilities

- Run descriptive statistics (.describe(), value counts) exclusively on the training data
- Create visual EDA plots with Matplotlib/Seaborn (e.g., age distribution vs. stroke risk, BMI vs. heart disease)
- Build a correlation heatmap for clinical features and flag extreme outlier vitals

Assigned deliverables

Comprehensive EDA notebook section with distribution plots; key medical risk insights passed to other members.

What I actually did

My part was the exploratory data analysis using the 1,400-row training set. I reviewed descriptive statistics and created Matplotlib charts showing the target class distribution, numeric feature distributions, stroke rates by age group and risk factors, and correlations between

features. I also checked BMI values and blood pressure readings. I found 10 BMI values below 15 and 124 rows where diastolic pressure was equal to or higher than systolic pressure. The feature distributions were not highly skewed, so a log transformation was not needed.

A surprising finding was that, in the training set, the rule "Age over 60, systolic blood pressure over 150, and smoker" matched all 71 stroke cases. This suggests that the target may have been generated using a simple rule, which limits what the model scores can tell us about real patients.

4. Great — Encoding, Scaling & Normalization Lead

Assigned responsibilities

- One-hot encode categorical features such as Gender with pd.get_dummies(drop_first=True)
- Apply StandardScaler to numerical vitals (BMI, blood pressure, cholesterol, glucose), strictly on the training data
- Normalize skewed distributions if necessary and print sample rows after each step

Assigned deliverables

Encoding and scaling code section; before/after feature scaling comparison plots; sample transformed rows output

What I actually did

I took the responsibility of designing and creating our data pipeline. My primary goal was to create a data infrastructure that was simple and reliable, and allowed my teammates to work on it efficiently. To this end, I implemented techniques to handle skewness, build out important clinical features and, most importantly, prevented data leakage. One of the most challenging tasks was the isolation of training and testing data. While using scalers or other power transformers on the training data, it is easy to have some testing data bleed into the training process. My solution was to build the pipeline so that every transformer fits only on the X_train before it even touches the X_test, with an additional step after each one to confirm its status. In addition to designing the main pipeline, I created and ran unit tests to make sure data processing didn't fail at any point. To speed the learning curve for my new team, I also documented the whole process. I designed and wrote the code for data processing in a more stable way to ensure the robustness of our baseline model.

5. Quan — Feature Engineering & Balancing Lead

Assigned responsibilities

- Engineer 2 to 4 clinical features (e.g., BP Risk Category, Pulse Pressure)
- Drop unnecessary identifier columns (Patient_ID)
- Analyze class imbalance for the target variable (Stroke)
- Apply resampling or class weighting (e.g., SMOTE) on the training set only

Assigned deliverables

Feature engineering code and rationale; before/after class balance plot; final cleaned and balanced training dataset export

What I actually did

I was responsible for feature engineering and class balancing. First, I dropped the Patient_ID column because it is only an identifier and could cause the model to memorize individual rows. Then I created new clinical features from the blood pressure, BMI, age, cholesterol and glucose columns, including Pulse Pressure, Mean Arterial Pressure, BMI Category, BP Risk Category and simple high-risk flags. I also analyzed the imbalance in the Stroke target, where only about 5% of patients had a stroke (about 71 positive cases against 1,329 negative in the training set). To fix this, I applied SMOTE-NC to the training set only, so the test set stayed untouched and kept its natural class ratio.
The main challenge was that Stroke is rare, so a model could look accurate while missing almost every stroke. I used SMOTE-NC instead of plain SMOTE because the data mixes continuous and 0/1 columns, and plain SMOTE can create impossible values like Smoking = 0.37. I also had to make sure balancing was fit on the training data only, so no information leaked from the test set.

6. Camila — Documentation & GitHub Lead

Assigned responsibilities

- Set up the GitHub repository structure and manage version control
- Write the 1-page project proposal / reflection journal and dataset URL documentation
- Compile individual member contribution logs
- Execute the Jupyter notebook end-to-end to ensure reproducibility and handle the final Canvas submission

Assigned deliverables

Complete GitHub repository;

1-page proposal PDF and reflection journal;

member contribution log;

final clean dataset (.csv)

Jupyter notebook (.ipynb)

What I actually did

My main contribution was organizing the project files and documentation. I organized the GitHub repository and brought together the Jupyter notebook, dataset source information, proposal, journals, and processed data files.
I prepared the one-page project proposal and documented the original dataset URL. I also collected and organized each team member's reflection and contribution entry. Before submitting, I reviewed the project materials against the assignment requirements, including the train/test split, EDA, preprocessing steps, plots, and exported datasets.
The notebook exports train_processed.csv, test_processed.csv, and train_balanced_smotenc.csv. My role helped keep the work organized so the team's code, results, and documentation could be reviewed together.